C Programming – Practice Question Bank (Topic-wise)

For MST Preparation | 2-Mark and 5-Mark Questions

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★ **Star marked questions are the most important from the exam and placement point of view. Practise these on priority.**

Disclaimer: These questions are for practice only. They are a helping guide for MST preparation and do not cover the complete syllabus, so do not rely on this list alone. Also study your textbook, class notes and solve additional problems.

Topic 1: Programming Fundamentals & Operators

2-Mark Questions

1. Define a variable and a constant with one example each.
2. List the basic data types in C with their typical sizes.
3. ★ What is the difference between the = and the == operator?
4. ★ Predict the output: int a=5, b=2; printf("%d %d", a/b, a%b);
5. What are format specifiers? Give any three with their use.
6. ★ Differentiate between ++i (pre-increment) and i++ (post-increment) with an example.

5-Mark Questions

1. Explain the different types of operators in C with examples.
2. ★ Write a program to swap two numbers (a) using a third variable and (b) without using a third variable.
3. Explain implicit and explicit type conversion (type casting) with examples.

Topic 2: Decision Making (if-else, switch)

2-Mark Questions

1. Write the syntax of the if-else-if ladder.
2. ★ Differentiate between the if-else statement and the switch statement.
3. What is a nested if? Give a short example.
4. ★ Predict the output: int x=3; if(x=0) printf("Zero"); else printf("Non-zero");
5. What is the use of the break statement inside a switch?

5-Mark Questions

1. ★ Write a program to find the largest of three numbers.
2. ★ Write a program using switch to make a simple calculator (+, -, *, /).
3. Write a program to check whether a given year is a leap year.

Topic 3: Loop Control Structures (important for MST)

2-Mark Questions

1. ★ Differentiate between the while loop and the do-while loop.

2. ★ Which loops are entry-controlled and which loop is exit-controlled?
3. ★ Write the syntax of a for loop and name its three parts.
4. ★ Predict the output: for(i=1;i<=5;i++){ if(i==3) continue; printf("%d ",i); }
5. ★ Differentiate between the break and the continue statement.
6. What is an infinite loop? Give an example.
7. ★ Predict the output: int i=0; while(i<5){ printf("%d ",i); i+=2; }
8. What is a nested loop? Where is it used?
9. Why is the goto statement generally avoided?

5-Mark Questions

1. ★ Write a program to print the sum of the first N natural numbers.
2. ★ Write a program to check whether a given number is prime.
3. ★ Write a program to print a pyramid / triangle star pattern.
4. ★ Write a program to reverse a number and check whether it is a palindrome.
5. ★ Write a program to find the factorial of a number using a loop.
6. ★ Write a program to print the Fibonacci series up to N terms.
7. Write a program to find the sum of digits of a number.
8. Explain break, continue and goto statements with examples.

Topic 4: Arrays (1D and 2D)

2-Mark Questions

1. Define an array. How is it declared in C?
2. ★ What is the index of the first and the last element of an array of size n?
3. How is an array stored in memory?
4. ★ Write the formula to calculate the address of element a[i].
5. How do you find the number of elements in an array using sizeof?
6. What is a two-dimensional array? Write its declaration.

5-Mark Questions

1. ★ Write a program to find the largest and smallest element in an array.
2. ★ Write a program to perform linear search in an array.
3. Write a program to find the sum and average of n numbers using an array.
4. ★ Write a program to add two matrices using 2D arrays.
5. Write a program to count the even and odd numbers in an array.

Topic 5: Strings

2-Mark Questions

1. What is a string in C? How does a string end?
2. ★ How much memory does the string "HELLO" take, and why?
3. ★ Name any four predefined string functions from string.h.
4. What does the strlen() function return?
5. Differentiate between gets() and scanf("%s", ...).

6. ★ How do you correctly compare two strings in C?

5-Mark Questions

1. ★ Write a program to find the length of a string without using strlen().
2. ★ Write a program to check whether a string is a palindrome.
3. Write a program to count the vowels and consonants in a string.
4. Write a program to reverse a string.
5. Explain any five string functions with examples.

Topic 6: Functions (including Recursion)

2-Mark Questions

1. What is a function? Why do we use functions?
2. Differentiate between a function declaration (prototype) and a function definition.
3. ★ Differentiate between call by value and call by reference.
4. What is a recursive function?
5. ★ What is the base condition in recursion, and why is it necessary?
6. Differentiate between built-in and user-defined functions.

5-Mark Questions

1. ★ Write a program to find the factorial of a number using recursion.
2. ★ Write a program to add two numbers using a user-defined function.
3. ★ Explain call by value and call by reference with separate programs.
4. Write a program to find the GCD of two numbers using recursion.
5. Write a program to print the Fibonacci series using recursion.

Note: Attempt the star-marked questions first, then the remaining ones. Write clean, well-commented programs and trace the output wherever asked.